

Wazuh Automated Security Alert Reporter

Home Lab Project Documentation

Sean Murphy | April 2026

1. Project Overview

This project extends an existing Wazuh SIEM home lab by building a Python-based automated security alert report generator. The script authenticates against the Wazuh REST API, queries the OpenSearch indexer for real-time alert data, and produces a formatted HTML report displaying timestamp, agent name, rule ID, alert description, and severity level for each event.

The project demonstrates practical skills in SIEM integration, REST API interaction, Python scripting, and security operations automation — built with the assistance of AI tooling to reflect real-world workflows where AI accelerates technical development.

2. Lab Environment

2.1 Infrastructure

- Host machine: HP Laptop — AMD Ryzen AI 5, 16GB RAM, Windows 11
- Hypervisor: Oracle VirtualBox
- Wazuh Server VM: Ubuntu 24.04 LTS, 6GB RAM allocated
- Monitored endpoint: Windows 11 (Sean_Murphy-PC) enrolled via NAT port forwarding
- Wazuh stack: Manager, OpenSearch Indexer, and Dashboard (all-in-one deployment)

2.2 Wazuh Dashboard — Environment Confirmed

With the VM running and all Wazuh services started, the dashboard confirmed one active agent connected and reporting in real time.

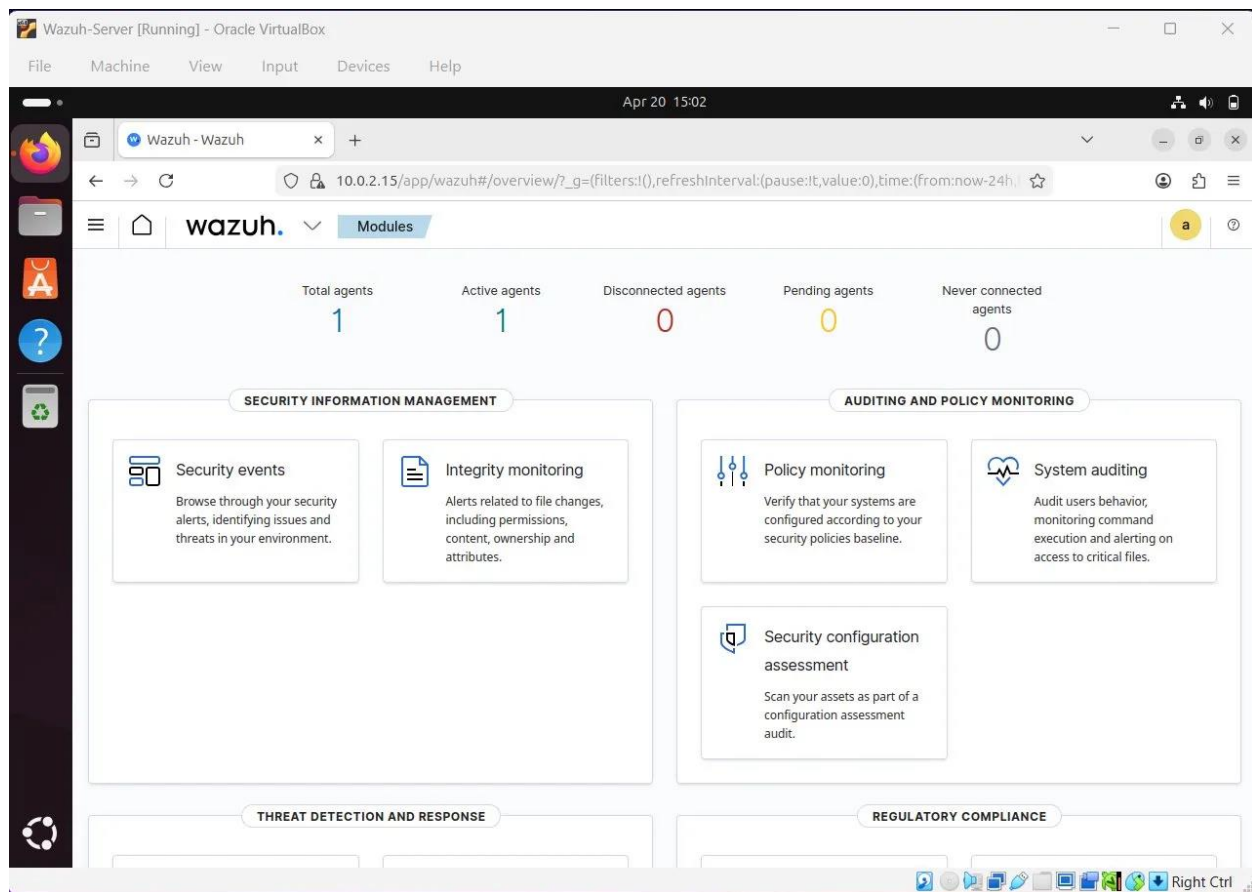


Figure 1 — Wazuh SIEM dashboard confirming the manager is online with one active Windows 11 endpoint agent connected and reporting.

3. API Authentication

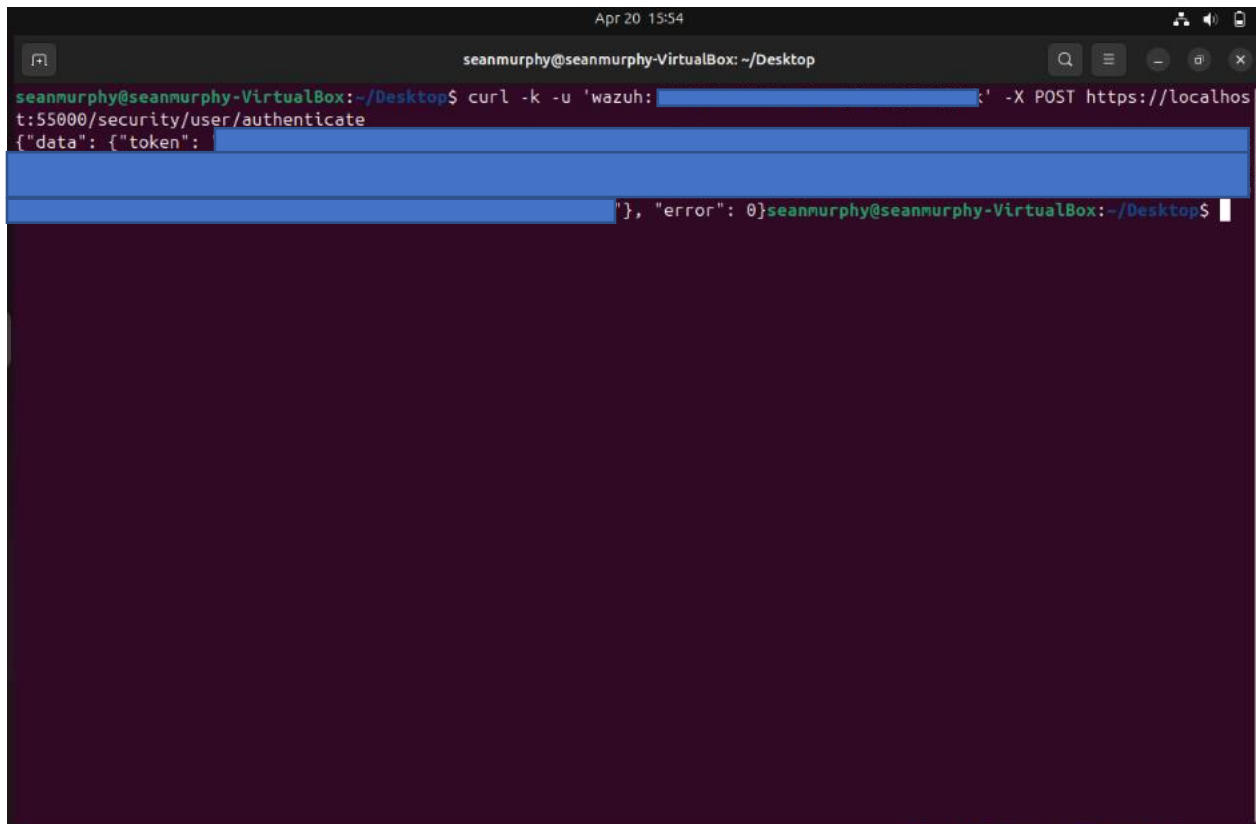
Wazuh exposes a REST API on port 55000. Before the Python script could query alert data, API credentials needed to be identified and validated. The default credentials had been changed during installation, requiring extraction of the original passwords from the installation archive.

3.1 Locating API Credentials

The Wazuh installation archive (wazuh-install-files.tar) stored on the VM desktop contained a passwords file listing all generated credentials for the manager, indexer, and API services. The file was extracted and read to identify the correct API username and password.

3.2 Validating API Access

Using cURL, the API was tested by requesting a JWT authentication token. A successful response confirmed the credentials were correct and the API was accessible.



```
seanmurphy@seanmurphy-VirtualBox: ~/Desktop
seanmurphy@seanmurphy-VirtualBox:~/Desktop$ curl -k -u 'wazuh:...' -X POST https://localhost:55000/security/user/authenticate
{"data": {"token": "..."}, "error": 0}seanmurphy@seanmurphy-VirtualBox:~/Desktop$
```

Figure 2 — Successful Wazuh API authentication via cURL, returning a JWT token confirming API access for the automated report script.

4. Script Development

The report generator was developed iteratively in Python 3, built directly on the Wazuh VM using the nano text editor. The script is composed of three functions:

- `get_token()` — authenticates against the Wazuh API and retrieves a JWT bearer token
- `get_alerts()` — queries the OpenSearch indexer directly on port 9200, retrieving the 50 most recent alerts sorted by timestamp
- `generate_report()` — parses each alert's fields and writes a formatted HTML table to the Desktop

4.1 Technical Approach

Alert data is stored in OpenSearch (the indexer component) rather than the Wazuh manager API directly. The script queries the `wazuh-alerts-*` index pattern using a POST request with a JSON body, authenticating with the OpenSearch admin credentials separately from the Wazuh API token. Each alert's `_source` object is parsed to extract the agent name, rule ID, rule description, and severity level.

4.2 Troubleshooting

Several issues were encountered and resolved during development, demonstrating the analytical problem-solving process required in real security operations environments:

- API credential mismatch — default credentials had been changed; resolved by extracting the passwords file from the installation archive
- OpenSearch query format — the alerts endpoint required a POST request with a JSON body rather than URL parameters; resolved by testing the raw API response structure
- HTML rendering issues — special characters in alert descriptions caused column misalignment; resolved by correcting malformed HTML tags in the report template
- Variable scoping error — a debug print statement referenced a variable before assignment; resolved by reordering the statements

4.3 Script Successfully Executing

After resolving all issues, the script ran successfully and confirmed the report was saved to the Desktop.

A screenshot of a Linux terminal window titled "seanmurphy@seanmurphy-VirtualBox: ~/Desktop". The terminal shows several commands being executed and their outputs:

```
alerts = get_alerts(token)
File "/home/seanmurphy/Desktop/wazuh-report.py", line 23, in get_alerts
    alerts = response.json()["data"]["affected_items"]
KeyError: 'data'
seanmurphy@seanmurphy-VirtualBox:~/Desktop$ nano -f /Desktop/wazuh-report.py
seanmurphy@seanmurphy-VirtualBox:~/Desktop$ python3 -f /Desktop/wazuh-report.py
File "/home/seanmurphy/Desktop/wazuh-report.py", line 22
    response = requests.get(url, headers=headers, params=params, auth=("admin", ))
SyntaxError: invalid syntax. Perhaps you forgot a comma?
seanmurphy@seanmurphy-VirtualBox:~/Desktop$ nano -f /Desktop/wazuh-report.py
seanmurphy@seanmurphy-VirtualBox:~/Desktop$ python3 -f /Desktop/wazuh-report.py
File "/home/seanmurphy/Desktop/wazuh-report.py", line 27
    alerts = [hit["_source"] for hit in hits]
              ^
SyntaxError: '[' was never closed
seanmurphy@seanmurphy-VirtualBox:~/Desktop$ nano -f /Desktop/wazuh-report.py
seanmurphy@seanmurphy-VirtualBox:~/Desktop$ python3 -f /Desktop/wazuh-report.py
/usr/lib/python3/dist-packages/urllib3/connectionpool.py:1100: InsecureRequestWarning: Unverified HTTPS request is being made to host 'localhost'. Adding certificate verification is strongly advised. See: https://urllib3.readthedocs.io/en/latest/advanced-usage.html#tls-warnings
warnings.warn(
<bound method Response.json of <Response [200]>>
/usr/lib/python3/dist-packages/urllib3/connectionpool.py:1100: InsecureRequestWarning: Unverified HTTPS request is being made to host 'localhost'. Adding certificate verification is strongly advised. See: https://urllib3.readthedocs.io/en/latest/advanced-usage.html#tls-warnings
warnings.warn(
Report saved to desktop as report.html
```


The last command prompt shown is "seanmurphy@seanmurphy-VirtualBox:~/Desktop\$", indicating the session continues.

Figure 3 — Terminal output confirming successful script execution. The script authenticated to the Wazuh API, queried the OpenSearch indexer, and saved the HTML report to the Desktop.

5. Report Output

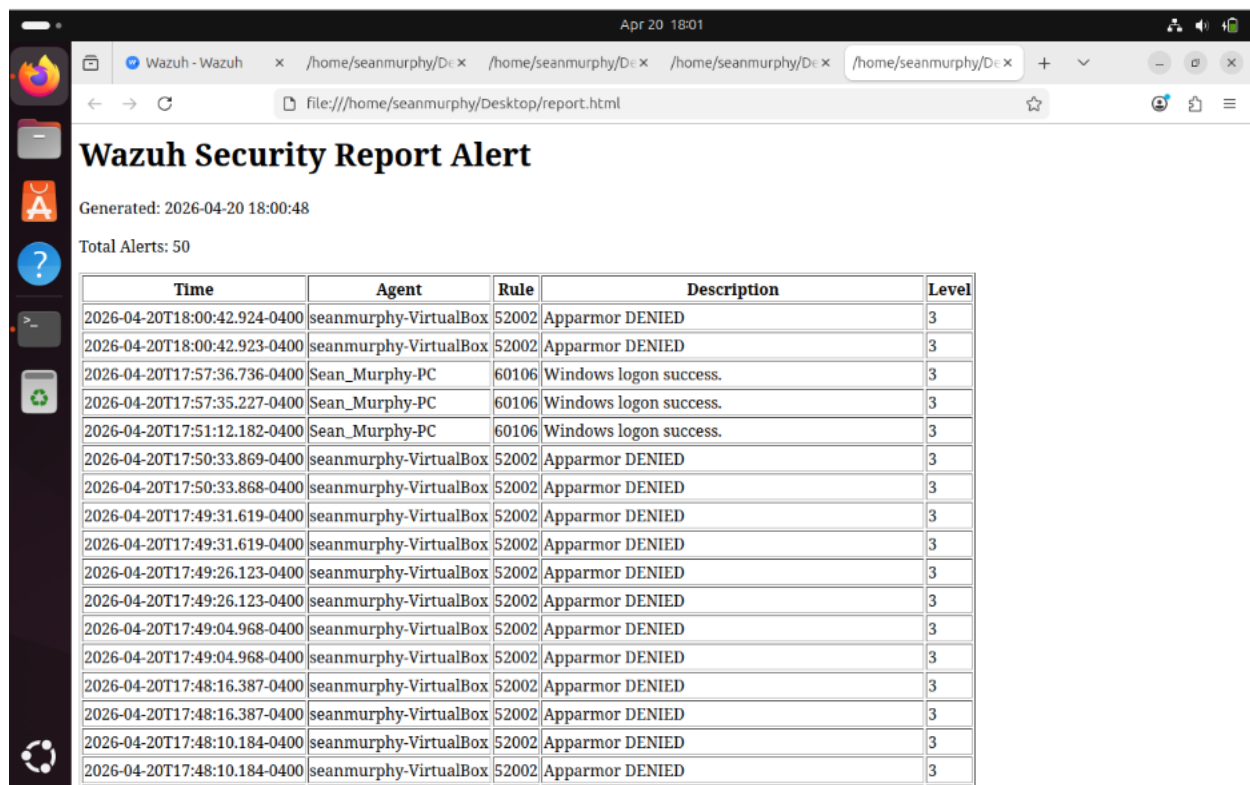
The generated HTML report displays 50 real-time security alerts pulled from the Wazuh OpenSearch indexer. Alerts are sourced from both the Ubuntu Wazuh server itself and the enrolled Windows 11 endpoint, demonstrating multi-agent monitoring capability.

5.1 Alert Data Captured

- Apparmor DENIED events — Linux kernel security module blocking unauthorized file access attempts on the Ubuntu server
- Windows logon success events — authentication events from the Windows 11 endpoint (Sean_Murphy-PC)
- Rule IDs 52002, 60106, 60642, 61104, 61138 — covering AppArmor, Windows authentication, and system events
- Severity level 3 across the majority of captured alerts

5.2 Completed Report

The final report rendered all five columns correctly — timestamp, agent name, rule ID, description, and severity level — with data from both monitored endpoints.



Wazuh Security Report Alert

Generated: 2026-04-20 18:00:48

Total Alerts: 50

Time	Agent	Rule	Description	Level
2026-04-20T18:00:42.924-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T18:00:42.923-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:57:36.736-0400	Sean_Murphy-PC	60106	Windows logon success.	3
2026-04-20T17:57:35.227-0400	Sean_Murphy-PC	60106	Windows logon success.	3
2026-04-20T17:51:12.182-0400	Sean_Murphy-PC	60106	Windows logon success.	3
2026-04-20T17:50:33.869-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:50:33.868-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:49:31.619-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:49:31.619-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:49:26.123-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:49:26.123-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:49:04.968-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:49:04.968-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:48:16.387-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:48:16.387-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:48:10.184-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3
2026-04-20T17:48:10.184-0400	seanmurphy-VirtualBox	52002	Apparmor DENIED	3

Figure 4 — Completed automated HTML security alert report showing 50 real-time alerts, displaying timestamp, agent name, rule ID, alert description, and severity level across both the Ubuntu server and Windows 11 endpoint.

6. Skills Demonstrated

- Wazuh SIEM deployment and administration — manager, indexer, and dashboard on Ubuntu 24.04
- REST API interaction — JWT token-based authentication against the Wazuh API
- OpenSearch querying — direct indexer queries using the search API with JSON request bodies
- Python scripting — building a functional automation tool using the requests library
- Multi-agent monitoring — capturing and reporting alerts from both Linux and Windows endpoints
- Iterative troubleshooting — identifying and resolving API, scripting, and rendering issues systematically
- AI-assisted development — using AI tooling to accelerate scripting, debug errors, and document the process

7. Tools and Technologies

- Wazuh SIEM (Manager, Indexer, Dashboard)
- OpenSearch REST API
- Python 3.12 with requests library
- Ubuntu 24.04 LTS
- Oracle VirtualBox
- Windows 11 (monitored agent endpoint)
- cURL (API testing and validation)

8. Project Repository

The complete script is available on GitHub with credentials removed and setup instructions included in the README.

github.com/murphys611/wazuh-alert-reporter